SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT

Spring 2024
Software Engineering (CS-226)
Assignment # 1

Semester: 4th | Batch: 2022F
Student's Name: Muhammad Faisal | Roll#: 2022F-BCS-152
Section: C

Total Marks: 4 | Marks Obtained:

CLO #	Course Learning Outcomes (CLOs)	PLO Mapping	Bloom's Taxonomy
CLO 1	**Describe** various software engineering processes and activities.	PLO_1 (Academic Education)	C2 (Understanding)

Instructions:

1. Use of Generative AI is not allowed.
2. Any answer that is more than 40% identical to another assignment's answer will get 0 marks.

Q1. Discuss how Technical Debt can be managed in a project.

Managing technical debt involves establishing processes and practices to prioritize, track, and mitigate its impact on software projects.

First, teams should create awareness and buy-in among stakeholders about the importance of addressing technical debt. This involves fostering a culture that values code quality, continuous improvement, and transparency about the trade-offs involved in software development.

Next, teams should prioritize technical debt alongside feature development and bug fixes, incorporating it into their regular sprint planning and backlog grooming sessions. This ensures that technical debt is not neglected or allowed to accumulate unchecked.

Furthermore, teams should establish clear criteria for identifying and categorizing technical debt, such as severity, impact on functionality, and estimated effort required for resolution. This allows for informed decision-making about which instances of technical debt to address and when.

Finally, teams should monitor and track technical debt over time, using metrics and tools to assess its impact on code quality, development velocity, and overall project health. By actively managing technical debt, teams can mitigate its negative consequences and maintain a sustainable pace of development.

Q2. Consider the following scenario.

"You and your team has worked hard on a project, the project's progress and feedback is really good". Every time you have delivered a feature of the project you have tested its working and asked client to verify the feature, the answer has been positive, But, one month before the project, the company director calls you and tells you that the client is not happy with the developed as it does not help them achieve their goals. While you were really confident at the start of the project that what you were developing was going to be really useful for the client, you now sit dejected in your office."

Discuss the most probable reason for this to occur? List down and describe the steps and activities within the Agile framework which are important to avoid such scenario.

The most probable reason for the scenario described is a lack of alignment between the developed product and the client's actual needs or goals. This misalignment could have occurred due to various reasons such as:

- Insufficient understanding of the client's requirements or goals at the beginning of the project.
- Changes in the client's priorities or business environment during the project lifecycle.
- Inadequate communication and feedback loops between the development team and the client.
- Failure to incorporate user feedback and validation throughout the development process.

To avoid such scenarios, Agile framework emphasizes the following steps and activities:

Continuous Collaboration: Regularly engage with the client to understand their evolving needs and priorities. This can be achieved through techniques like daily stand-up meetings, sprint reviews, and retrospectives.

Iterative Development: Break down the project into small, incremental deliverables that can be regularly reviewed and validated by the client. This allows for early detection of misalignment and enables course corrections as needed.

User Stories and Acceptance Criteria: Define clear user stories and acceptance criteria in collaboration with the client. This ensures that the development efforts are focused on delivering value that aligns with the client's goals.

Adaptability: Embrace change and be responsive to feedback throughout the project lifecycle. Agile frameworks like Scrum and Kanban provide mechanisms for incorporating changes without disrupting the overall progress.

Empowered Teams: Empower cross-functional teams to make decisions and adapt to evolving requirements. This fosters a culture of ownership and accountability, where team members are invested in delivering outcomes that meet the client's needs.

Q3. Refer to the document at: https://pdf.usaid.gov/pdf_docs/Pnacu897.pdf Identify if this documentation was created using knowledge of Software Engineering or Software Project Management. Defend your answer.

The document provided appears to be related to the field of Software Project Management rather than Software Engineering. It focuses on the management and monitoring of projects funded by the United States Agency for International Development (USAID), outlining procedures, guidelines, and reporting requirements for project implementation. While software engineering principles may be applied within the context of managing software projects, the document primarily addresses project management aspects such as planning, monitoring, evaluation, and reporting, which are central to Software Project Management.

Q4. Explain why an incremental software development model is beneficial to the project's client.

An incremental software development model is beneficial to the project's client for several reasons:

Early Delivery of Value: Incremental development allows for the delivery of working software in small increments. This means that the client can start realizing benefits and value from the software earlier in the project lifecycle, rather than waiting for the entire project to be completed.

Flexibility and Adaptability: Incremental development enables the client to provide feedback and make changes throughout the development process. This flexibility allows for the incorporation of new requirements, adjustments, and improvements based on evolving needs and market conditions.

Risk Mitigation: By delivering software incrementally, the project risk is spread out over multiple iterations. This reduces the likelihood of large-scale project failures and provides opportunities to identify and address issues early in the development process.

Enhanced Transparency: Incremental development promotes transparency and collaboration between the development team and the client. Regular demonstrations and reviews of completed increments allow the client to track progress, provide feedback, and ensure alignment with their expectations.

Improved Quality: Incremental development encourages frequent testing and validation of software components. This iterative approach to development helps in identifying and addressing defects early, resulting in higher overall software quality.

Overall, an incremental software development model offers the client greater control, visibility, and value throughout the project lifecycle, leading to increased satisfaction and successful outcomes.